Dobre Andreea Viktorya
AAB master

Laboratory 3 FDSAI

Ex 1: Screen time over the week

The first exercise visualises the number of hours spent in front of a screen on each day of the week (Monday to Sunday). The data were entered as a pandas DataFrame with two columns, Day and ScreenTime, and all charts were produced with Matplotlib.

Day	Mo	Tu	We	Th	Fr	Sa	Su
Hours	5	6	4	7	8	10	9

	Day	ScreenTime
0	Mo	5
1	Tu	6
2	We	4
3	Th	7
4	Fr	8
5	Sa	10
6	Su	9

a) + b) Line chart with title, labelled axes, units and grid

The line chart shows the weekly pattern. It has a title, labelled axes with the unit (hours) stated on the vertical axis, markers on every observation and the value written above each point. The vertical axis starts at 0 so that the heights are not exaggerated, and the top and right spines were removed to reduce visual clutter.

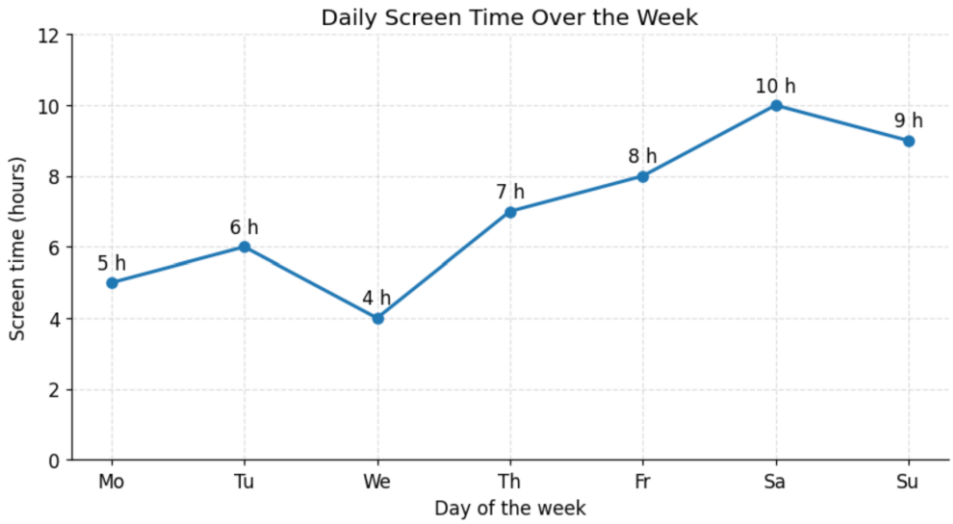


Figure 1. Daily screen time over the week (line chart)

The grid was tested by plotting the same chart with and without it. A light, dashed, semi-transparent grid helps the reader estimate values against the axis, while a heavy grid would compete with the data line. Because every point is also labelled directly, the grid is useful but optional.

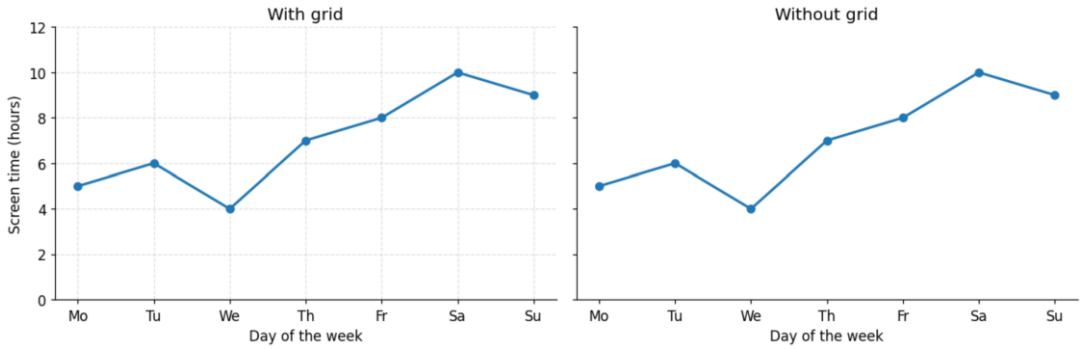


Figure 2. Comparison of the line chart with and without grid

c) Bar chart version

The second version uses a bar chart with the same title, axis labels, unit and value labels. The bars start at zero (mandatory for bar charts) and the weekend days (Saturday and Sunday) are highlighted in a different colour to support the comparison between weekdays and weekend.

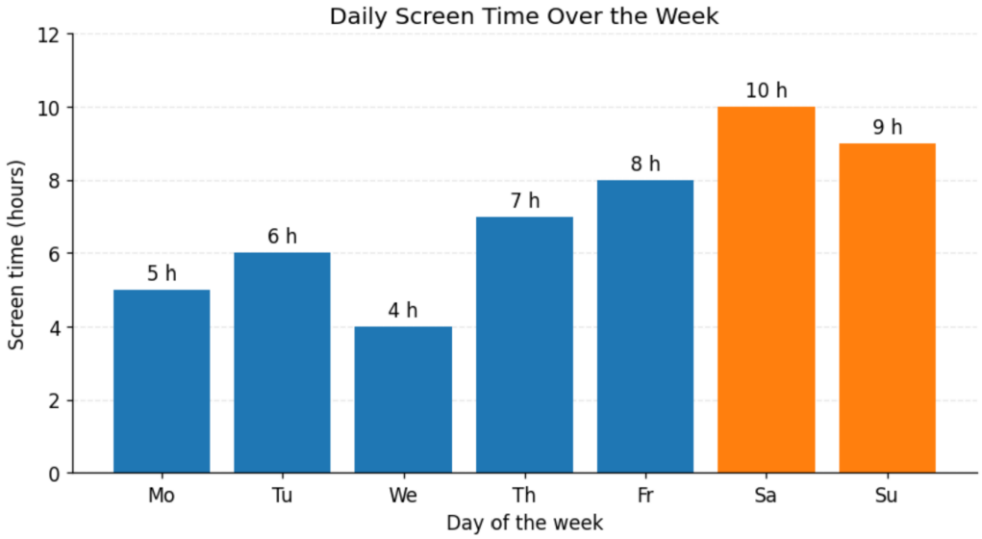


Figure 3. Daily screen time over the week (bar chart)

Numerical summary

Statistic	**Value**	**Comment**
Weekly mean	7.0 h	49 hours in total over 7 days
Weekday mean (Mo–Fr)	6.0 h	30 hours over 5 days
Weekend mean (Sa–Su)	9.5 h	19 hours over 2 days
Minimum / Maximum	4 h / 10 h	Wednesday / Saturday

Interpretation

Screen time is lowest on Wednesday (4 hours), then rises almost steadily to a peak of 10 hours on Saturday and drops slightly on Sunday (9 hours). The weekend average (9.5 hours) is about 60% higher than the weekday average (6 hours).

Dobre Andreea Viktorya
AAB master

Which chart communicates the pattern more clearly? The line chart is better for the pattern itself: the days form an ordered sequence, and the slope between consecutive points makes the dip on Wednesday and the climb towards the weekend immediately visible. The bar chart is better for comparing the exact values of individual days (for example Saturday is 2.5 times Wednesday) and, with the colour highlight, for contrasting weekdays and weekend. Since the question is about the pattern over the week, the line chart communicates it more clearly, while the bar chart is a good complement for reading individual values.

Ex 2: Plotting functions

Five functions were plotted over the interval [−6, 6], using 1000 equally spaced points so that the curves are smooth.

The following best practices were applied: one function per subplot with its own title, labelled axes (x and f(x)), black reference lines through the origin, a light dashed grid, a dense sampling of x, different colours for different functions, and an additional chart where functions with a similar output range are compared on the same axes with a legend.

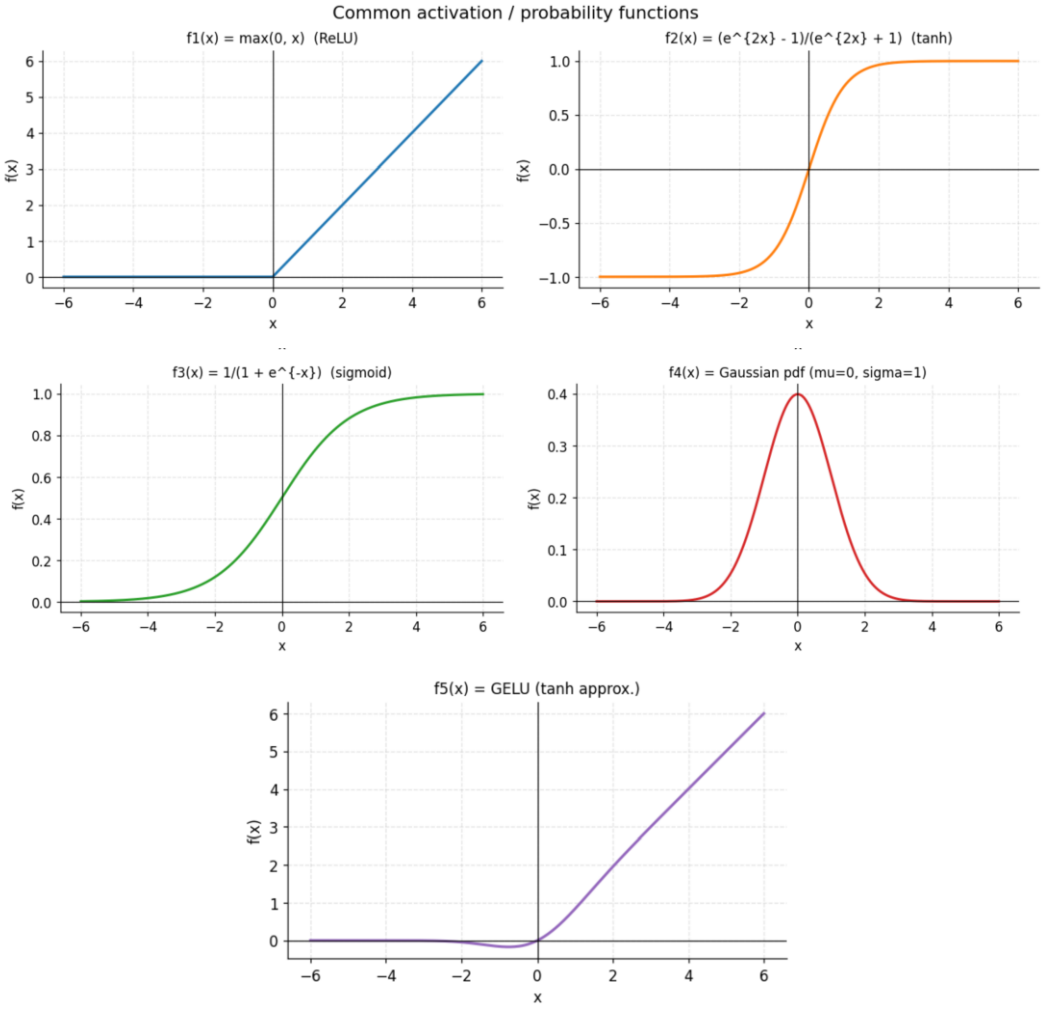


Figure 4. Functions f1–f4: ReLU, tanh, sigmoid and Gaussian density, f5: GELU (tanh approximation)

Dobre Andreea Viktorya
AAB master

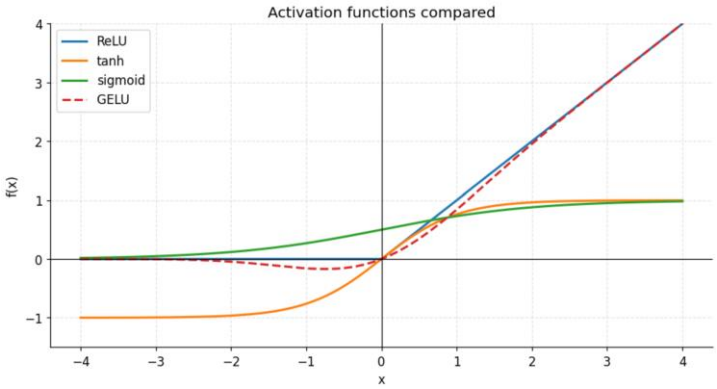


Figure 6. ReLU, tanh, sigmoid and GELU on the same axes

Interpretation

The functions differ mainly in their output range and symmetry:

- ReLU is zero for negative inputs and equal to x for positive ones, so its range is [0, ∞) and it has a sharp corner at the origin.
- tanh is S-shaped, zero-centred and bounded between −1 and 1, while the sigmoid has the same shape but is bounded between 0 and 1 and equals 0.5 at x = 0. Both saturate for large |x|.
- The Gaussian density is bell-shaped and symmetric around 0, with a maximum of about 0.4 at x = 0 and values close to zero beyond |x| = 3.
- GELU behaves like a smooth version of ReLU: it is almost zero for large negative x, dips slightly below zero (minimum of about −0.17 near x = −0.75) and approaches the identity for large positive x.

In the comparison chart, ReLU and GELU overlap for large positive x, while tanh and sigmoid saturate at 1. The Gaussian density is plotted separately, since its much smaller range (maximum of about 0.4) would make it hard to read on the same axes.**Ex 3: Student scores and outliers**

The data contain the scores of 10 students.

Student	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Score	70	75	78	82	74	90	92	95	60	100

	Student	Score
0	1	70
1	2	75
2	3	78
3	4	82
4	5	74
5	6	90
6	7	92
7	8	95
8	9	60
9	10	100

Dobre Andreea Viktorya
AAB master

a) Boxplot to identify possible outliers

The boxplot was drawn with Matplotlib, and potential outliers would be shown as red circles beyond the whiskers. The outlier rule used is the IQR rule: a value is an outlier if it lies below Q1 − 1.5·IQR or above Q3 + 1.5·IQR.

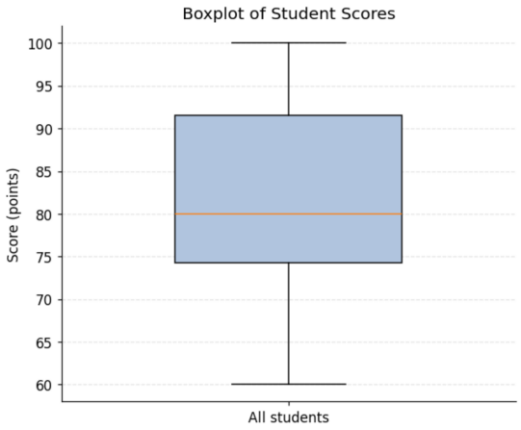


Figure 7. Boxplot of the student scores

```
Q1 = 74.25, Q3 = 91.5, IQR = 17.25
Lower fence = 48.38, Upper fence = 117.38

Outliers found:
None - all scores are inside the fences.
```

b) Scatter plot with outliers in a separate colour

In the scatter plot, normal values are blue and any outlier would be drawn in red. The dotted lines are the IQR fences. Although no point is an outlier under the IQR rule, the two most extreme values (60 and 100) are marked with orange rings to show which points are the most distant from the rest.

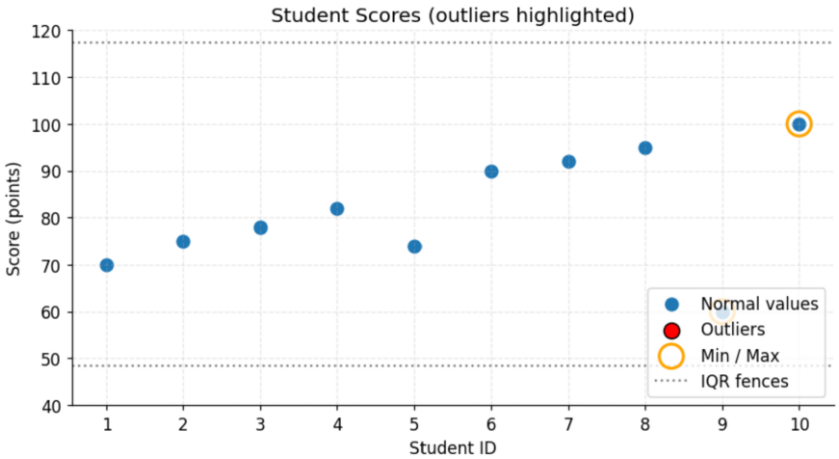


Figure 8. Scatter plot of the scores with the IQR fences

Interpretation

No outliers are detected: the lowest score (60) and the highest score (100) are both inside the interval [48.38, 117.38]. The distribution is slightly spread out (standard deviation about 12.5 points) and the median (80) is close to the mean (81.6), so there is no sign of a strongly atypical score.

Dobre Andreea Viktorya
AAB master

c) How can misleading axis scales make outliers seem “normal”?

Outliers are judged visually by their distance from the other points, and the axis scale controls that distance. The same data were plotted with three different vertical ranges, and an extra experiment adds an artificial value of 400 to compare a linear and a logarithmic axis.

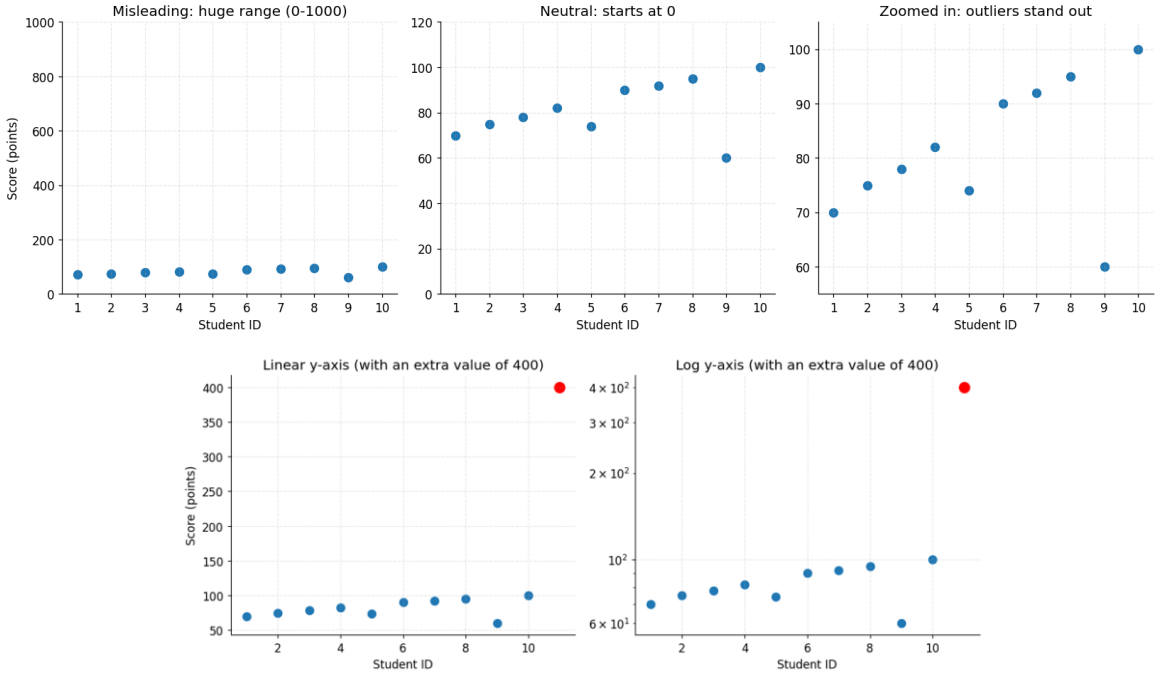


Figure 9. The same scores with three different y-axis ranges & Linear versus logarithmic y-axis with an artificial extreme value (400)

With a very wide range (0–1000) all scores are squeezed into a thin band at the bottom and even the lowest and highest values look just like the others. With a range that starts at 0 the picture is neutral, and with a zoomed-in range (55–105) the differences are exaggerated. A logarithmic scale compresses large values, so an extreme value such as 400 appears much closer to the rest than it really is.

Best practice: use an honest and clearly labelled scale (start bars at zero, use a reasonable range for scatter plots, state explicitly when a log scale is used) and do not rely on the picture alone. A boxplot and numerical criteria such as the IQR fences or z-scores should support the visual conclusion.